

Omar Elhanafy

US Citizen | Lake Forest, California, United States | oelhanafy12@gmail.com

+1 714-342-7476 | linkedin.com/in/omar-elhanafy-9a9257211 | github.com/omargalal20

Experienced Software Engineer building production-ready microservices, cloud infrastructures, and AI integrated systems

Work Experience

AI Software Engineer Intern, Product Manager Accelerator Feb 2025 - April 2025

- Worked on a voice-first, AI-powered EdTech interactive storytelling platform for children that is educational, ethical, and engaging.
- Selected as **team lead** to guide **3 developers**, offering mentorship on backend development, code reviews, and Agile best practices, resulting in accelerating the delivery time of the **MVP**
- Broke down user stories into technical tasks and collaborated with product managers on **sprint planning** and **scope definition** in an **Agile Environment**.
- Designed and authored the **Software Design Document**, architecting the foundation for cloud, system, and component architectures.
- Built **APIs** using **FastAPI** and **MongoDB**; developed **RBAC-based security middleware** to manage user access.
- Accelerated development through **CI/CD pipelines** using **GitHub Actions**, deployed on **Google Cloud Run**, with configurations for **WebSockets** and **AI** and **LLM** model latency.
- Integrated **Speech-to-Text**, **LLM**, and **Text-to-Speech** utilizing **Langchain** to deliver an end-to-end voice-to-voice chat.

Software Engineer, Progressio Solutions July 2023 – Aug 2024

- Optimized infrastructure implementation time by **50%** by building reusable infrastructures and pipelines in **Terraform** and **Jenkins** for both **AWS** and **Azure**.
- Reduced infrastructure **costs** in **AWS** by **30%** by identifying underutilized services, and introducing instance right-sizing.
- Automated the creation of **Kubernetes** and **Docker Swarm** in **Terraform**.
- Incorporated **IAM** best practices in Terraform deployments, enhancing **cloud security** posture across multiple environments.
- Migrated company's main server from **Linode** to **Azure** using **Terraform** to automate the infrastructure creation.
- Developed of **5+ RESTful microservices** utilizing **Kotlin**, **Spring Boot**, and **PostgreSQL**.
- Built reusable microservices (Authentication, Authorization, File Upload) integrated with **AWS**, enabling scalable integration across multiple internal projects and reducing development time by **40%**.
- Led a client project and accelerated the delivery by **30%** by facilitating collaboration with the project manager, designers, and developers on client requirements.
- Collaborated with cross-functional teams and clients to define requirements, conduct code reviews, and ensure delivery of features aligned with product goals in a **Scrum** environment.

Junior ReactJS Developer, ShopX Nov 2022 – March 2023

- Implemented the front end of an e-commerce website using **ReactJS**.
- Utilized **Redux** for state management and adhered to best practices.
- Containerized and deployed the frontend on **AWS**, reducing deployment complexity and ensuring high availability.

Data Analyst Research Intern, German University In Cairo July 2022 – October 2022

- Worked on a transformation pipeline for supply chain tabular data to graph data for graph algorithms to be used in **Neo4j**
- Automated data extraction and processing using web scraping in **Selenium** to retrieve real-time transportation costs, reducing manual effort by **30%**
- Utilized **NLP** for centralizing column name matching for different input tables for data preprocessing.

Data Structures And Algorithms Junior Teaching Assistant, German University In Cairo Oct 2021 – Jan 2022

- Assigned to two Teaching Assistants and assisted in teaching 2 separate classes of 2nd-year undergraduate students.
- Assisted students in applying **data structures and algorithms** in **Java**

Education

California State University, Long Beach Master of Science in Computer Science	Part-time, Expected July 2027
German University in Cairo Bachelor of Science in Computer Science and Engineering Bachelor Thesis: Implemented and compared Machine Learning and Deep Learning models to detect heart arrhythmia from a wearable sensor in real-time, achieving 90% accuracy.	Sept. 2018 – July 2023
University of California, Irvine Visiting Scholar, Research Exchange	March 2022 – June 2022
German International University in Berlin Semester Abroad	Sept. 2019 – Jan. 2020

Projects

Medical Q&A Assistant - Built a healthcare-focused Q&A system leveraging React, FastAPI, LangChain, LangGraph, FHIR, and Gemini to provide real-time, general and patient-specific medical insights to healthcare professionals. - Automated GCP infrastructure provisioning with Terraform, improving deployment speed by 50%.	Project URL
Supply Chain Data Transformation Pipeline Implemented in Python a transformation pipeline for supply chain tabular data into graph representations in Neo4j to improve queryability and enable route optimization using Neo4j algorithms.	Github Repo

Certifications

AWS Certified Cloud Practitioner	Certificate Link
Machine Learning Specialization - Deeplearning.AI	Certificate Link
LangChain - Develop LLM powered applications with LangChain	Certificate Link

Skills

Programming Languages: Java, Kotlin, Python, Javascript, Typescript, HCL

Technologies: Terraform, Spring Boot, FastAPI, Nginx, Traefik, Keras, Numpy, Pandas, Scikit-learn, Nodejs, Linux, Supabase, Selenium, FHIR, HL7

Cloud and CI/CD: AWS, S3, Cognito, Azure, GCP, Identity Platform, Cloud Run, Linode, Docker, Kubernetes, Jenkins, Github Actions, Git, GitHub, Jira, Azure DevOps

AI/ML: Machine Learning, Neural Networks, Large Language Models, LangChain, LangGraph, LangSmith, Google Gemini, Universal-2, OpenAI (GPT, Whisper), Deep Learning, Model Training and Evaluation

Databases: MySQL, Postgres, Neo4j, MongoDB, Firebase

Other: System Architecture, Software Design, Software Documentation, Microservices, Object Oriented Programming, Functional Programming, Agile, Confluence

Languages: English, Arabic